

Lab: 3 Secure Data Communication

1. Introduction

This lab demonstrates the most basic form of data communication: connecting two PCs directly with a cable. Students will assign IP addresses to each PC and verify communication using the ping command.

2. Learning Objectives

By the end of this lab students will be able to:

1. Connect two PCs directly using a cable.
2. Assign static IP addresses to PCs.
3. Verify direct communication using the ping command.

3. Topology

[PC1] <---- Crossover Cable ----> [PC2]

4. IP Addressing Table

Device	IP Address	Subnet Mask
PC1	192.168.1.10	255.255.255.0
PC2	192.168.1.20	255.255.255.0

5. Step-by-step Implementation

Step 1 — Connect devices

- Take two PCs in Packet Tracer (or real PCs).
- Connect them directly using a **crossover cable** (if using old NICs) or a straight-through cable (modern NICs auto-sense).

Step 2 — Configure IP addresses

- On **PC1**:
 - IP Address: 192.168.1.10 ◦ Subnet Mask: 255.255.255.0
- On **PC2**:
 - IP Address: 192.168.1.20 ◦ Subnet Mask: 255.255.255.0

(No default gateway needed since there is no router.)

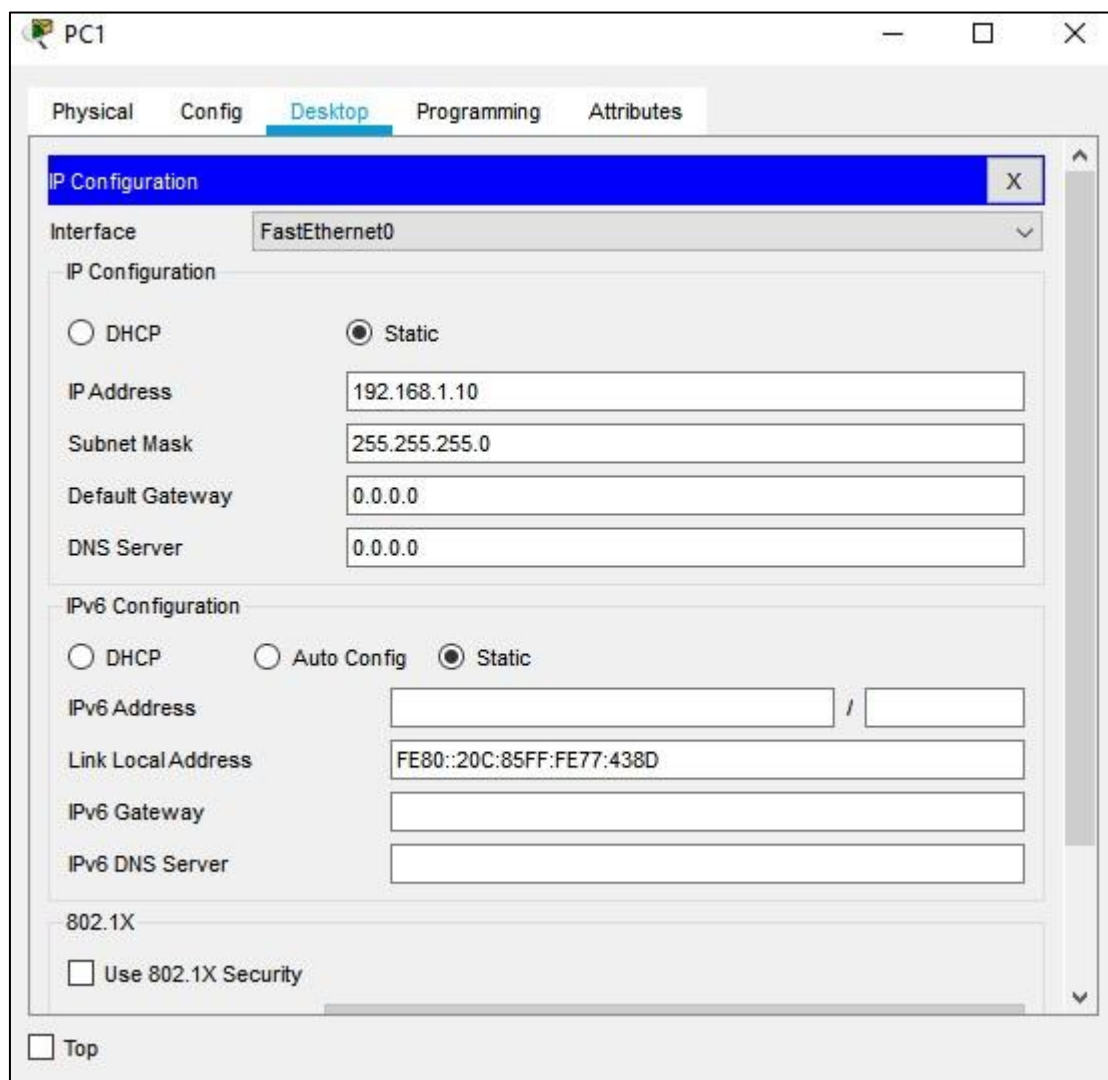
Step 3 — Test connectivity

- On **PC1**, open Command Prompt and type:
- ping 192.168.1.20
- On **PC2**, ping back to PC1:
- ping 192.168.1.10

If configured correctly, both PCs will successfully exchange ICMP echo replies.

Lab Task (Attach Screenshots)

1. Connect two PCs directly with a cable.
2. Assign IP addresses as shown in the table.
3. Use ping to test communication between the two PCs.
4. Submit screenshots of:
 - PC1 and PC2 IP configurations.
 - Successful ping results in both directions.



PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.1.20

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:FFFF:FEBC:AE86

IPv6 Gateway

IPv6 DNS Server

802.1X

☐ Use 802.1X Security

☐ Top

